B1GMB42 ZFS Recovery

B1GMB42 ZFS Recovery — rpool and bpool from live media

Machine: Dell Precision T5810 (Tower5810)
Installed pools: `rpool` (root + `/home`) on Hitachi `sdb4` + TEAM `sda` special vdev; `bpool` (`/boot`) on Hitachi `sdb2`
Boot environment: `rpool/ROOT/ubuntu_cortt9`, `bpool/BOOT/ubuntu_cortt9`
Encryption: off (no ZFS passphrase on these pools)

Use this when the installed Ubuntu system will not boot. Boot **Ventoy Ubuntu 26.04** from **Wiggly** (`sdc1`) instead — do not import `rpool` until you are ready to repair.

Required host setting: force import

On the **installed** system, `/etc/default/zfs` must include:

```
ZPOOL_IMPORT_OPTS="-f"
```

Without this, boot can fail or hang after a recovery `zpool export`, hostid mismatch, or unclean shutdown — initramfs import will not force-import the pools. Recovery scripts already pass `-f` on the command line; the installed OS does **not** unless this default is set.

```
# verify on Tower5810 (or inside chroot after mount)
grep '^ZPOOL_IMPORT_OPTS' /etc/default/zfs
# expect: ZPOOL_IMPORT_OPTS="-f"
```

One-shot alternative: add `zfsforce=1` on the GRUB/kernel command line for a single boot.

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1. Choose your path

Situation	What to use
Ventoy live, IndianaDell or DOSBOOT scripts available	**Section 2** (recommended)
Ventoy live, no scripts — only a shell	**Section 3** (manual `zpool`)
Installed system still boots but you need a read-only view	`mount-rpool-recovery.sh mount --overlay`

DOSBOOT copy: `IndianaDell/recovery/` on partition **DOSBOOT** (`sdc3`, vfat).
Repo copy: `~/Documents/IndianaDell/mount-rpool-recovery.sh` and `scripts/recovery/`.

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2. With IndianaDell / DOSBOOT scripts

Prerequisites (live session)

```
sudo apt-get update
sudo apt-get install -y zfsutils-linux
```

Mount rpool for chroot (default)

```
cd /path/to/recovery/scripts    # DOSBOOT/IndianaDell/recovery or IndianaDell repo
sudo bash mount-rpool-recovery.sh mount      # use bash on DOSBOOT (vfat)
sudo bash mount-bpool-recovery.sh mount      # puts /boot at /recovery/boot
sudo bash mount-rpool-recovery.sh status
```

You should see `/recovery/bin`, `/recovery/home`, `/recovery/etc`, and `/recovery/boot` (kernel + initrd).

Enter chroot and repair

```
sudo ./mount-rpool-recovery.sh chroot
# inside chroot:
mount | grep boot
zpool status
apt-get update
apt-get install -y --reinstall zfs-initramfs grub-efi-amd64-signed shim-signed
update-initramfs -c -k all
update-grub
exit
```

Tear down

```
sudo ./mount-bpool-recovery.sh umount
sudo ./mount-rpool-recovery.sh umount
```

Overlay fallback (running system already on rpool)

Only when a full live-media chroot is impossible:

```
sudo ./mount-rpool-recovery.sh mount --overlay
ls /recovery/home/user
sudo ./mount-rpool-recovery.sh umount
```

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3. Without scripts (manual commands)

Run as **root** from Ventoy Ubuntu live. Replace dataset names if yours differ (`zpool import` / `zfs list` to discover).

3a. Import and mount rpool

```
sudo mkdir -p /recovery
sudo zpool import -N -f -R /recovery -d /dev/disk/by-id rpool
sudo zfs mount -a -R /recovery
sudo zfs list -r rpool | head -20
```

Boot environment should appear under `/recovery` (e.g. `/recovery/home/user`).

3b. Import and mount bpool at /recovery/boot

```
sudo zpool import -N -f -d /dev/disk/by-id bpool
BOOT_DS=$(zfs list -H -o name,mountpoint -r bpool/BOOT | awk '$2=="/boot"{print $
sudo zfs set mountpoint=/recovery/boot "$BOOT_DS"
sudo zfs mount "$BOOT_DS"
ls /recovery/boot
```

3c. Chroot

```
for d in dev proc sys run; do sudo mount --bind /$d /recovery/$d; done
sudo mount --bind /dev/pts /recovery/dev/pts
sudo chroot /recovery /bin/bash
```

Inside chroot: run repairs (Section 4), then `exit`.

3d. Unmount and export

```
sudo zfs umount -a -R /recovery
sudo zpool set altroot=- rpool
sudo zpool export rpool
sudo zfs umount "$BOOT_DS"
sudo zpool export bpool
```

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4. Common repairs (inside chroot)

Problem	Commands
Broken initramfs / boot	`update-initramfs -c -k all` && `update-grub`
Boot hangs on ZFS import after recovery	Ensure `/etc/default/zfs` has `ZPOOL_IMPORT_OPTS="-f"`; rebuild initramfs if needed
Pool won't import (live)	`zpool import -f -d /dev/disk/by-id` (inspect candidates); check `dmesg`
Pool degraded	`zpool status -v`; replace faulted vdev per ZFS docs
PERC H710 fault	SAS bays unavailable — see hardware manual; pool on SATA only
Boot menu missing	`apt-get install --reinstall grub-efi-amd64-signed shim-signed`
ZFS mount fails	`zfs mount -a`; check `zfs list -o name,mountpoint,canmount`

Before leaving chroot, confirm force import is set:

```
grep '^ZPOOL_IMPORT_OPTS' /etc/default/zfs
# must show: ZPOOL_IMPORT_OPTS="-f"
# if missing or empty:
#   printf 'ZPOOL_IMPORT_OPTS="-f"\n' | tee -a /etc/default/zfs   # or edit in pla
# then: update-initramfs -c -k all
```

Do not re-enable ZFS encryption until TPM + recovery are documented (hardware manual).

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5. Disk map (this machine)

Device	Pool / role
`sdb4`	`rpool` main vdev
`sda`	`rpool` special vdev (TEAM SSD)
`sdb2`	`bpool`
`sdb1`	EFI (`/boot/efi` when running)
`sdb3`	Plain **4 GiB** swap (HDD)
`rpool/swap`	**33 GiB** swap zvol (prefers special/SSD; fstab `pri=10,nofail`)
`sdc1` Wiggly	Ventoy + Ubuntu live ISO + persistence
`sdc3` DOSBOOT	Recovery scripts + this manual

Before `zpool export rpool`: disable the zvol swap so export is clean:

```
sudo swapoff /dev/zvol/rpool/swap 2>/dev/null || sudo swapoff -a
# HDD swap sdb3 can stay or also swapoff -a
```

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6. Script reference

Script	Purpose
`mount-rpool-recovery.sh`	Import rpool at `/recovery`, chroot, overlay, umount
`mount-bpool-recovery.sh`	Import `bpool`, mount boot FS at `/recovery/boot`

Environment overrides: `POOL_NAME`, `BPOOL_NAME`, `RECOVERY_ROOT`.

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7. Verify after repair

Reboot to installed disk (not Ventoy). Then:

```
grep '^ZPOOL_IMPORT_OPTS' /etc/default/zfs    # must be "-f"
zpool status rpool bpool
zfs list -r rpool | head -15
ls /boot/grub
```

If boot fails again, repeat from Section 2 or 3. If import stalls at boot, boot Ventoy, set `ZPOOL_IMPORT_OPTS="-f"` in the chroot (Section 4), update-initramfs, and retry.

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